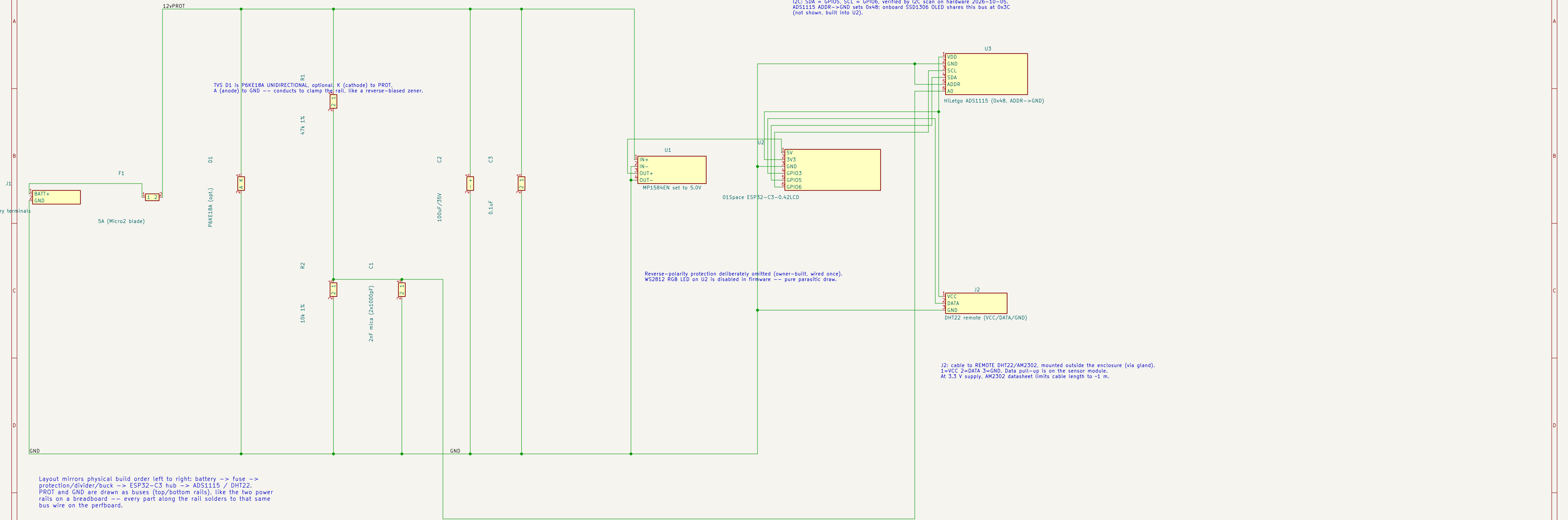


EV 12V BATTERY MONITOR
Battery -> WiFi -> HomeAssistant



Layout mirrors physical build order left to right: battery -> fuse -> protection/divider/buck -> ESP32-C3 hub -> ADS1115 / DHT22. PROT and GND are drawn as buses (top/bottom rails), like the two power rails on a breadboard -- every part along the rail solders to that same bus wire on the perfboard.

Divider R1 47k / R2 10k = 5.7:1 (unit 1 measured 46.5k/9.85k = 5.72:1). C1 = 2 x 1000pF mica in parallel (2nF, low leakage). ADS1115 abs max = VDD+0.3 = 3.6V. 12.22V(Hooded 50% SOC)->2.14V 14.4V(charging)->2.52V 16.0V(design max)->2.80V @ A0. ANALOG GROUND: R2 and C1 return to the ADS1115 GND pin, and that node meets power GND at ONE point. Load current through a shared ground path adds directly to A0 and is scaled up 5.7x (bench: 15mV rail-to-rail = ~86mV error at the battery).

See CLAUDE.md / PARTS.md in project root for full design rationale
ESP32-C3 + ADS1115 battery monitor. Prototype: 2027 Subaru Uncharted GT
Owner-built

Sheet: /
File: irin.kicad_sch

Title: Irin - 12V Battery Monitor

Size: A2	Date: 2026-08-29	Rev: B
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